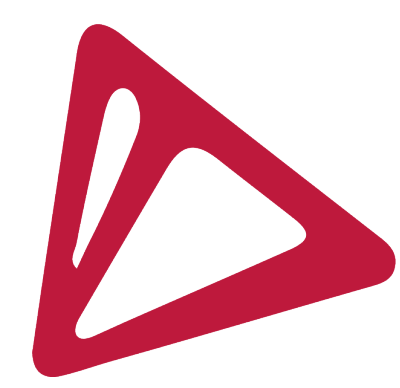


3D Dental Crown Generation using Geometry Images and Latent Diffusion

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SFU

60x

LOWER RECONSTRUCTION ERROR

First **geometry-image** representation for dental crowns

0.733

BITE PROXIMITY · 127 INTERSECTIONS — BOTH BEST

First **latent diffusion** model for crown generation

11,513

ANNOTATED SCANS, RELEASED

FDI 16 dataset scans paired with surrounding dentition

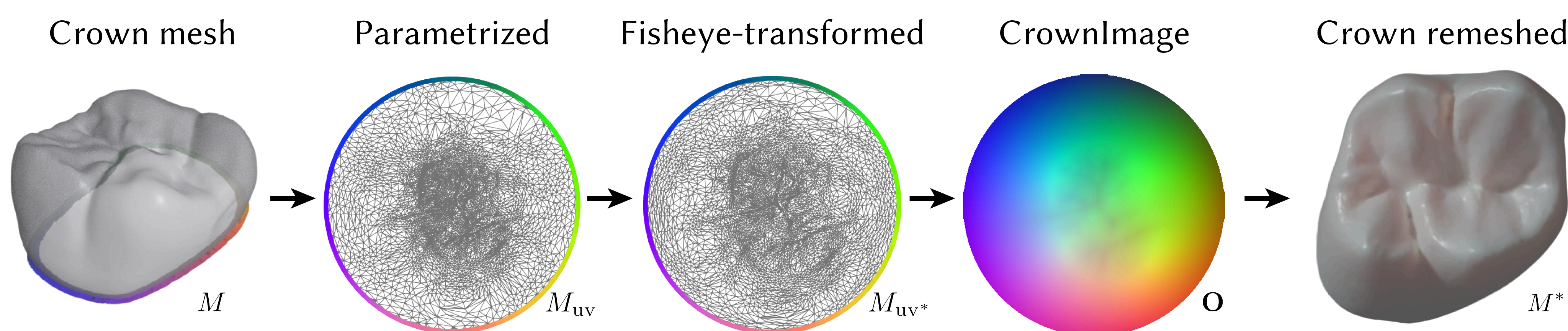
Crown design is labour-intensive and admits **multiple valid solutions** per patient, a natural fit for **generative** modelling. Every candidate must still **occlude correctly** with the opposing dentition and carry the natural fissures that guide food flow. But the shape representation has been the bottleneck: existing choices force a tradeoff between full 3D detail and 2D generative tooling, leaving no representation that supports both.

Point clouds smooth away the sharp fissures that define an occlusal surface, and still need a separate meshing stage.

Depth images project from a single viewpoint, discarding occluded regions like the lingual wall, or else demand multi-view fusion, adding registration complexity to the task.

Key insight. A crown is a dome with a rim, so it flattens onto a disk with no cuts and no seams. Each pixel stores its surface point's 3D position: the crown becomes an ordinary 2D image, a **geometry image**, and the mesh is rebuilt from it exactly.

Pixel color encodes 3D position (**R** = x, **G** = y, **B** = z): the rainbow disk is the crown surface itself, not a texture.



Flatten the mesh M onto the unit disk with a boundary-fixed harmonic map.

A radial warp spends resolution on the occlusal centre and saves it at the axial rim.

Sample the inverse position map onto an $H \times W \times 3$ grid; each pixel stores (x, y, z) .

Triangulate adjacent occupied pixels to rebuild M^* , no marching cubes, no post-hoc surfacing.

The diagram illustrates the FLUX model architecture, divided into two main sections: **Reconstruction** and **Generation**.

Reconstruction: This section shows a process where a latent representation (a sphere with a rainbow gradient and a gray object) is processed by a **FLUX Encoder** (indicated by a flame icon) and then a **FLUX Decoder** (indicated by a flame icon) to reconstruct the original latent representation.

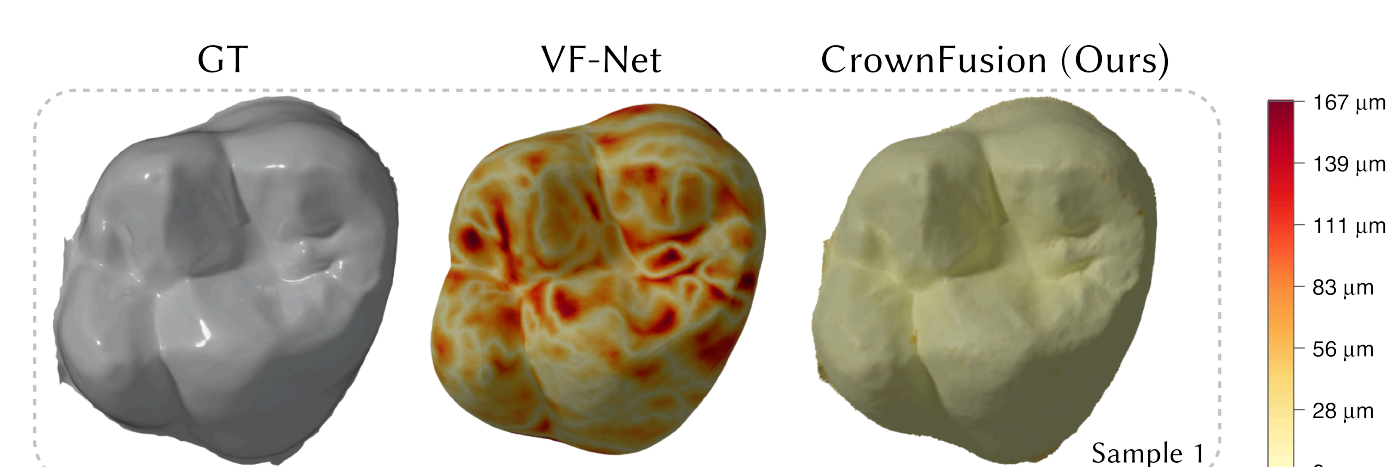
Generation: This section shows a process where a latent representation (a sphere with a rainbow gradient and a gray object) is processed by a **SiT Diffusion model** (indicated by a flame icon) and then a **FLUX Decoder** (indicated by a snowflake icon) to generate a new latent representation.

Encodes CrownImages into compact latents. Finetuning matters: geometry images carry **coordinates**, not colour, so their distribution is far from FLUX's pretraining data. A small latent error is a **visible geometric distortion** on the reconstructed surface.

A Scalable Interpolant Transformer trained from scratch in that latent space, cross-attended on the neighbouring and antagonist teeth via a jointly trained 3DShape2VecSet encoder, denoising towards a valid crown.

Generation is bounded by autoencoder fidelity. CrownImages reconstruct **two orders of magnitude** more accurately than point-cloud autoencoders, which blur the occlusal fissures and round off the **margin line**.

	CD $\times 10^2 \downarrow$	EMD $\times 10^2 \downarrow$
LION	5.35	22.85
FoldingNet	5.26	33.67
VF-Net	1.21	6.30
CrownFusion	0.02	0.45



	CD-L2↓	Bite↓	Int. #↓
DMC	0.433	0.734	256
VBOD	0.353	0.747	143
CrownFusion	0.412	0.733	127
ground truth	—	0.716	12

Bite proximity, distance to the antagonist, CrownFusion lands closest to ground truth and **halves** DMC's intersections.

Each crown is a draw from the same conditioning, same neighbours, same antagonist, different noise, a shortlist of admissible crowns, not one averaged answer. Samples vary in fissure pattern, secondary fossae and oblique ridge, where clinicians have design freedom, while outline and ridge height stay stable so each still meets its neighbours.